

Aspects of Non-lattice Formalism of Supersymmetric Field Theories

SUMMER PROJECT REPORT



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Abstract

The statistical mechanics of a quantum harmonic oscillator can easily be formulated in terms of Feynman path integrals in imaginary time. By straightforward integration one can find the partition function, internal energy and correlation functions. When time is discretized, the same results can be obtained directly using Monte Carlo methods on a computer. In the zero-temperature limit one recovers the usual quantum mechanical results.

In this project we looked at the of path integral formalism of quantum harmonic oscillator, and non-lattice formalism of a supersymmetric quantum mechanics with symmetric and non-symmetric superpotentials.

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1 Quantum Harmonic Oscillator

1.1 Introduction

Feynman's formulation of quantum mechanics in terms of path integrals arise from obscurity in the literature to the forefront of today's research in statistical physics and quantum field theory. Finite temperature properties of a quantum mechanical system can be found by the method of functional integration (Feynman's path integral formalism).

This formalism is applied in Sec. 1.3 to the harmonic oscillator where all quantum fluctuations can be obtained directly. If the zero temperature limit is taken one recovers the standard quantum mechanical results involving only the ground state.

The following sections are based on Ref. [1].

1.2 Finite temperature quantum mechanics

For a quantum mechanical system with Hamiltonian

$$H = \frac{p^2}{2m} + V(q), \quad (1)$$

in thermal equilibrium at temperature T , the partition function is

$$Z = \text{Tr} \exp(-\beta H), \quad (2)$$

with $\beta = 1/kT$, k is the Boltzmann's constant.

In the coordinate basis with $\langle q'|q \rangle = \delta(q' - q)$ the trace is simply the integral

$$Z(\beta) = \int_{-\infty}^{\infty} dq \langle q | \exp(-\beta H) | q \rangle. \quad (3)$$

Comparing the operator $\exp(-\beta H)$ with the time evolution operator $\exp(itH/\hbar)$ we see that the matrix element in Eq. (3) gives the transition amplitude for the particle to move from position q and back again to q in an imaginary time interval

$$t = i\beta\hbar.$$

In order to write Eq. (3) as a functional integral, we partition the time interval $\beta\hbar$ into N segments, each of duration a so that

$$\beta\hbar = Na.$$

Inserting the complete set of coordinate eigenstates, corresponding to each of these short time intervals, the original matrix element can be written as a product of N transition amplitude of the form

$$\langle q' | \exp(-aH/\hbar) | q \rangle.$$

For very short lattice distances, a , one finds, for the partition function

$$Z(\beta) = \int Dq \exp(-A[q]/\hbar). \quad (4)$$

Here the integration measure

$$\int Dq = C^N \prod_{n=1}^N \int_{-\infty}^{\infty} d(q_n) \quad (5)$$

and the action

$$A[q] = \sum_{n=1}^N a \left[\frac{m}{2a^2} (q_{n+1} - q_n)^2 + V(q_n) \right]. \quad (6)$$

Because of the trace in Eq. (2), we now have the periodic boundary condition

$$q_{N+1} = q_1.$$

In the limit $a \rightarrow 0$ and $N \rightarrow \infty$, with β fixed, the action Eq. (6) takes the general integral form

$$A[q] = \int_0^{\beta\hbar} d\tau \left[\frac{m}{2} \dot{q}^2 + V(q) \right], \quad (7)$$

where τ is imaginary or Euclidean time equal to Na when a is finite.

From the partition function $Z(\beta)$ we directly obtain the free energy $F(\beta)$ from $Z = \exp(-\beta F)$. The internal energy $U(\beta)$ is the expectation value of the kinetic energy $T = -\frac{m}{2}\dot{q}^2$ plus the potential energy $V(q)$. From the virial theorem one has

$$\langle T \rangle = \frac{1}{2} \langle qV' \rangle, \quad (8)$$

for bounded motion.

Expectation values are now calculated as in ordinary statistical mechanics. For example, the average potential energy is

$$\langle V \rangle = \frac{1}{Z} \int Dq V(q) \exp(-A/\hbar). \quad (9)$$

Instead of using the virial theorem one can directly obtain the kinetic energy from the expectation value of the time-split product of the squared velocity

$$\langle T \rangle = \frac{-m}{2a^2} \langle (q_{n+1} - q_n)(q_n - q_{n-1}) \rangle, \quad (10)$$

where the different terms are given by the correlation function

$$C_{n-m} = \langle q_n q_m \rangle.$$

In the zero temperature limit $\beta \rightarrow \infty$ the internal energy V equals the ground state energy E_0 .

1.3 The harmonic oscillator

The path integral Eq. (4) for the partition function can be written as a product of simple Gaussian integrals for the harmonic oscillator with the potential

$$V(q) = \frac{1}{2} m \omega^2 q^2.$$

This is done by introducing the discrete Fourier components u_k of the position coordinates

$$q_n = N^{-1/2} \sum_{k=1}^N u_k \exp^{i \frac{2\pi}{N} kn}. \quad (11)$$

Using the completeness relation

$$\sum_{n=1}^N \exp^{i \frac{2\pi}{N} (k-k')n} = N \delta_{kk'}, \quad (12)$$

we also have the inverse transform

$$u_k = N^{-1/2} \sum_{n=1}^N q_n \exp^{-i \frac{2\pi}{N} kn}.$$

Since the coordinates q_n are real we have the conditions

$$u'_k = u_{-k},$$

which makes the number of independent real Fourier components equal to N .

Now

$$\sum_{k=1}^N q_n^2 = \sum_{n=1}^N u'_k u_k,$$

and

$$\sum_{k=1}^N (q_{n+1} - q_n)^2 = \sum_{k=1}^N 2u'_k u_k (1 - \cos \frac{2\pi}{N} k),$$

so that the action Eq. (6) becomes diagonal

$$A[q] = \frac{m}{2a} \sum_{n=1}^N u'_k \left[2(1 - \cos \frac{2\pi}{N} k) + a^2 \omega^2 \right] u_k. \quad (13)$$

The integration measure Eq. (5) is now

$$\int Dq = \Delta \int Du = \Delta C^N \prod_{k=1}^N \int_{-\infty}^{\infty} du_k, \quad (14)$$

where the Jacobian

$$\Delta = |\partial q_n / \partial u_k|.$$

Using Eq. (11) and Eq. (12) one finds

$$\Delta \Delta' = 1.$$

That is,

$$\Delta = 1.$$

1.3.1 Partition function

The net result of the integration can be written as

$$Z(\beta) = (\det M)^{-1/2}, \quad (15)$$

with

$$\det M = \prod_{k=1}^N \left[2(1 - \cos \frac{2\pi}{N} k) + a^2 \omega^2 \right], \quad (16)$$

setting

$$\cosh x = 1 + \frac{1}{2} a^2 \omega^2,$$

and

$$R = e^{-x} = 1 + \frac{1}{2}a^2\omega^2 - a\omega(1 + \frac{1}{4}a^2\omega^2)^{\frac{1}{2}}, \quad (17)$$

the product in the determinant Eq. (16) can be simplified using the identity

$$\prod_{k=1}^N \left(\cosh x - \cos \frac{2\pi}{N} k \right) = 2^{1-N} (\cosh Nx - 1). \quad (18)$$

The above relations give us the following partition function

$$Z(\beta) = \frac{R^{N/2}}{1 - R^N}. \quad (19)$$

1.3.2 Derivation of the partition function

$$\begin{aligned}
 Z(\beta) &= (\det M)^{-1/2} \\
 &= \frac{1}{\prod_{k=1}^N [2(1 - \cos(\frac{2\pi}{N}k)) + a^2\omega^2]} \\
 &= \frac{1}{2^N \prod_{k=1}^N [(1 - \cos(\frac{2\pi}{N}k)) + \frac{a^2\omega^2}{2}]} \\
 &= \frac{1}{2^N \prod_{k=1}^N [(1 - \cos(\frac{2\pi}{N}k)) + (\cosh x - 1)]} \\
 &= \frac{1}{2^N \prod_{k=1}^N [\cosh x - \cos(\frac{2\pi}{N}k)]} \\
 &= \frac{1}{2(\cosh Nx - 1)} \quad \text{(using identity)} \\
 &= \frac{1}{2(\frac{\exp^{Nx} + \exp^{-Nx}}{2} - 1)} \\
 &= \frac{1}{(R^{-N} + R^N - 2)^{1/2}} \\
 &= \frac{1}{R^{-N/2} - R^{N/2}} \\
 &= \frac{1}{R^{-N/2}[1 - R^N]} \\
 &= \frac{R^{N/2}}{1 - R^N}
 \end{aligned}$$

1.3.3 Correlation function

We have the expression for the correlation function

$$C_s = \langle q_{n+s} q_n \rangle$$

Using Eq. (11) we see that its Fourier transform is just the expectation value

$$\langle u'_k u_k \rangle = \frac{a\hbar}{m} \frac{\delta_{kk'}}{2(1 - \cos(\frac{2\pi}{N}k) + a^2\omega^2)},$$

which gives

$$C_s = \frac{a\hbar}{2Nm} \sum_{k=1}^N \frac{\cos(\frac{2\pi}{N}ks)}{\cosh x - \cos(\frac{2\pi}{N}k)}. \quad (20)$$

Simplifying we get

$$C_s = \frac{\hbar}{2m\Omega} \frac{R^s + R^{N-s}}{1 - R^N}, \quad (21)$$

where

$$\Omega^2 = \omega^2(1 + \frac{1}{4}a^2\omega^2).$$

2 Non-lattice Formalism of Supersymmetric Quantum Mechanics

This section is based on Ref. [2].

2.1 Introduction

The supersymmetric quantum mechanics model is defined by the action [2]

$$S = \int_0^\beta dt \left[\frac{1}{2} (\partial_t \phi)^2 + h'(\phi)^2 + \bar{\psi} (\partial_t + h''(\phi)) \psi \right], \quad (22)$$

where ϕ is a scalar field, and ψ is a one-component Dirac field, both in one dimension, obeying periodic boundary conditions. This model is supersymmetric for the function (called the superpotential)

$$h(\phi) = \frac{1}{2} m \phi^2 + \frac{1}{4} g \phi^4,$$

and supersymmetry will break for the function

$$h(\phi) = g(\phi) \left(\frac{1}{3} \phi^2 + m^2 \right). \quad (23)$$

In our approach we make a Fourier expansion

$$\phi(t) = \sum_{n=-\Lambda}^{\Lambda} \tilde{\phi}_n \exp^{in\omega t}; \quad \omega = \frac{2\pi}{\beta}, \quad (24)$$

and similarly for the fermionic fields, where n takes integer values, and Λ is the UV cutoff. In terms of the Fourier modes, the action S can be written as

$$S = S_B + S_F,$$

where S_B is the bosonic part of the action and S_F the fermionic part of the action.

2.2 Analysis for symmetric superpotential

We have

$$h(\phi) = \frac{1}{2} m \phi^2 + \frac{1}{4} g \phi^4. \quad (25)$$

The bosonic part of the action is

$$S_B = \beta \left[\sum_{n=-\Lambda}^{\Lambda} \frac{1}{2} \{ (n\omega)^2 + m^2 \} \tilde{\phi}_n \tilde{\phi}_{-n} + mg(\tilde{\phi}^4)_0 + \frac{1}{2} g^2 (\tilde{\phi}^6)_0 \right], \quad (26)$$

and the fermionic part is

$$S_F = \sum_{nk} \tilde{\psi}_n M_{nk} \tilde{\psi}_k \quad (27)$$

where the fermionic operator is

$$M_{nk} = \beta \left[(in\omega + m) \delta_{nk} + 3g(\tilde{\phi}^2)_l \delta_{n,k+l} \right]. \quad (28)$$

We have the derivative of the superpotential

$$h'(\phi) = \frac{\partial}{\partial \phi} \left(\frac{1}{2} m \phi^2 + \frac{1}{4} g \phi^4 \right) = m \phi + g \phi^3 \quad (29)$$

and the second derivative

$$h''(\phi) = \frac{\partial}{\partial \phi} (m \phi + g \phi^3) = m + 3g \phi^2. \quad (30)$$

2.2.1 Derivation of the bosonic action

We have the bosonic action

$$S_B = \int_0^\beta dt \left[\frac{1}{2} (\partial_t \phi)^2 + h'(\phi)^2 \right]. \quad (31)$$

The time derivative of the field is

$$\partial_t \phi = \frac{\partial}{\partial t} \phi(t) = \sum_{n=-\Lambda}^{\Lambda} i\omega n \tilde{\phi}_n e^{i\omega n t}. \quad (32)$$

The square of it is

$$(\partial_t \phi)^2 = \left[\sum_{n=-\Lambda}^{\Lambda} i\omega n \tilde{\phi}_n e^{i\omega n t} \right] \left[\sum_{n=-\Lambda}^{\Lambda} i\omega n \tilde{\phi}_n e^{i\omega n t} \right]. \quad (33)$$

Simplifying

$$(\partial_t \phi)^2 = -\omega^2 \sum_m \sum_n m n \tilde{\phi}_m \tilde{\phi}_n e^{i\omega(m+n)t}. \quad (34)$$

Performing the integration

$$\begin{aligned}
 \int_0^\beta dt \frac{1}{2} (\partial_t \phi)^2 &= \frac{1}{2} \omega^2 \sum_{m,n} mn \tilde{\phi}_m \tilde{\phi}_n \int_0^\beta dt e^{i\omega(m+n)t} \\
 &= \frac{1}{2} \omega \sum_{m,n} -mn \tilde{\phi}_m \tilde{\phi}_n 2\pi \delta(m+n) \\
 &= \frac{1}{2} 2\pi \omega \sum_{m,n} -mn \tilde{\phi}_m \delta(m - (-n)) \\
 &= \frac{1}{2} 2\pi \omega \sum_n (n) n \tilde{\phi}_{-n} \tilde{\phi}_n \\
 &= \frac{1}{2} \beta \sum_n \frac{2\pi}{\beta} \omega n^2 \tilde{\phi}_{-n} \tilde{\phi}_n \\
 &= \beta \sum_n \frac{1}{2} (\omega n)^2 \tilde{\phi}_{-n} \tilde{\phi}_n.
 \end{aligned} \tag{35}$$

We have

$$\begin{aligned}
 \int_0^\beta dt \frac{1}{2} (h'(\phi))^2 &= \int_0^\beta dt \frac{1}{2} (m\phi + g(\phi)^3)^2 \\
 &= \int_0^\beta dt \frac{1}{2} [m^2 \phi^2 + g^2 \phi^6 + 2mg\phi^4],
 \end{aligned}$$

$$\begin{aligned}
 \int_0^\beta dt \frac{1}{2} (m^2 \phi^2) &= \frac{m^2}{2} \int_0^\beta dt \phi^2 \\
 &= \frac{m^2}{2} \int_0^\beta dt \tilde{\phi}_m \exp^{i\omega m t} \tilde{\phi}_n \exp^{i\omega n t} \\
 &= \frac{m^2}{2} \int_0^\beta dt \tilde{\phi}_m \tilde{\phi}_n \exp^{i\omega(m+n)t} \\
 &= \frac{2\pi m^2}{2} \tilde{\phi}_m \tilde{\phi}_n \delta(\omega(m+n)) \\
 &= \frac{2\pi m^2}{\omega 2} \tilde{\phi}_m \tilde{\phi}_n \delta((m+n)) \\
 &= \frac{\beta m^2}{2} \tilde{\phi}_{-n} \tilde{\phi}_n,
 \end{aligned}$$

$$\begin{aligned}
 \int_0^\beta dt \, mg\phi^4 &= mg \int_0^\beta dt \sum_{a,b,c,d} \tilde{\phi}_a \tilde{\phi}_b \tilde{\phi}_c \tilde{\phi}_d e^{i\omega(a+b+c+d)t} \\
 &= \frac{2\pi mg}{\omega} \sum_{a,b,c,d} \tilde{\phi}_a \tilde{\phi}_b \tilde{\phi}_c \tilde{\phi}_d \delta(a+b+c+d) \\
 &= \beta mg \sum_{b,c,d} \tilde{\phi}_{-b-c-d} \tilde{\phi}_b \tilde{\phi}_c \tilde{\phi}_d \\
 &= \beta mg (\tilde{\phi}^4)_0
 \end{aligned}$$

$$\begin{aligned}
 \int_0^\beta dt \, g^2 \phi^6 &= \frac{g^2}{2} \int_0^\beta dt \sum_{a,b,c,d,e,f} \tilde{\phi}_a \tilde{\phi}_b \tilde{\phi}_c \tilde{\phi}_d \tilde{\phi}_e \tilde{\phi}_f e^{i\omega(a+b+c+d+e+f)t} \\
 &= \frac{g^2}{2} \sum_{a,b,c,d,e,f} \tilde{\phi}_a \tilde{\phi}_b \tilde{\phi}_c \tilde{\phi}_d \tilde{\phi}_e \tilde{\phi}_f \delta(a+b+c+d+e+f) \\
 &= \frac{g^2}{2} \sum_{b,c,d,e,f} \tilde{\phi}_{-b-c-d-e-f} \tilde{\phi}_b \tilde{\phi}_c \tilde{\phi}_d \tilde{\phi}_e \tilde{\phi}_f \\
 &= \frac{g^2}{2} (\tilde{\phi}^6)_0.
 \end{aligned}$$

2.2.2 Fermionic part of the action

We have the fermionic part of the action

$$S_F = \int_0^\beta dt \bar{\psi} (\partial_t + h''(\phi)) \psi. \quad (36)$$

$$\begin{aligned}
 \int_0^\beta dt \, \bar{\psi} \partial_t \psi &= \int_0^\beta dt \sum_n (\tilde{\psi}_n) e^{i\omega n t} \sum_k i\omega k \tilde{\psi}_k e^{i\omega k t} \\
 &= \sum_{n,k} i\omega k \tilde{\psi}_n \tilde{\psi}_k \int_0^\beta dt e^{i\omega(n+k)t} \\
 &= 2\pi \sum_{n,k} i\omega k \tilde{\psi}_n \tilde{\psi}_k \delta(\omega(n+k)) \\
 &= \frac{2\pi}{\omega} \sum_{n,k} i\omega k \tilde{\psi}_n \tilde{\psi}_k \delta(n+k) \\
 &= \sum_{n,k} \tilde{\psi}_n \beta (ik\omega) \delta_{nk} \tilde{\psi}_k,
 \end{aligned}$$

and

$$\begin{aligned}
 \int_0^\beta dt \bar{\psi}(m + 3g\phi^2)\psi &= \int_0^\beta dt \sum_{a,b} m \tilde{\psi}_a \tilde{\psi}_b e^{i\omega(a+b)t} \\
 &\quad + 3g \int_0^\beta dt \sum_{a,b,c,d} \tilde{\psi}_a \tilde{\phi}_b \tilde{\phi}_c \tilde{\psi}_d e^{i\omega(a+b+c+d)t} \\
 &= \sum_{n,k} \tilde{\psi}_n \beta (m\delta_{nk} + 3g(\tilde{\phi}^2)_l \delta_{n,k+l}) \tilde{\psi}_k.
 \end{aligned}$$

The above two integrals, when combined, gives

$$S_F = \sum_{nk} \tilde{\psi}_n M_{nk} \tilde{\psi}_k, \quad (37)$$

where

$$M_{nk} = \beta [(in\omega + m)\delta_{nk} + 3g(\tilde{\phi}^2)_l \delta_{n,k+l}]. \quad (38)$$

Integrating out the fermions first, we obtain the effective action for the bosons as

$$S_{eff} = S_B - \ln \det M. \quad (39)$$

2.2.3 Derivation of effective action for bosons

We have

$$\begin{aligned}
 \int d\phi d\bar{\psi} d\psi \exp(S_B + S_F) &= \int d\phi d\bar{\psi} d\psi \exp(-S_B) \exp(-S_F) \\
 &= \int d\phi e^{-S_B} \det M \\
 &= \int d\phi e^{-S_B} \exp(\ln \det M) \\
 &= \int d\phi \exp(-S_B + \ln \det M) \\
 &= \int d\phi \exp(-[S_B - \ln \det M]).
 \end{aligned}$$

We have the gradient of the effective action

$$\begin{aligned} \frac{\partial S_{eff}}{\partial \tilde{\phi}_n} = & \beta \left[\{(n\omega)^2 + m^2\} \tilde{\phi}_{-n} + 4mg(\tilde{\phi})_{-n}^3 + 3g^2(\tilde{\phi}^5)_{-n} \right. \\ & \left. - Tr \left(\frac{\partial M}{\partial \tilde{\phi}_n} M^{-1} \right) \right]. \end{aligned} \quad (40)$$

Note that the gradient of the action is needed for Monte Carlo simulations.

2.3 Analysis of non-symmetric superpotential

We have the superpotential

$$h(\phi) = g\phi \left(\frac{1}{3}\phi^2 + m^2 \right). \quad (41)$$

Using similar calculation techniques as used for previous form of the potential we can write the action again as the sum of bosonic and fermionic parts.

We have

$$h'(\phi) = g(\phi^2 + m^2), \quad (42)$$

and

$$h''(\phi) = 2g\phi. \quad (43)$$

The bosonic part of the action is

$$S_B = \beta \left[\frac{1}{2}g^2m^4 + \sum_{n=-\delta}^{\delta} \frac{1}{2} \{ (n\omega)^2 + 2g^2m^2 \} \tilde{\phi}_{-n} \tilde{\phi}_n + \frac{g^2}{2} (\phi^4)_0 \right]. \quad (44)$$

The fermionic part is

$$S_F = \sum_{nk} \tilde{\psi}_n M_{nk} \tilde{\psi}_k, \quad (45)$$

where the fermionic operator is

$$M_{nk} = \beta \left[(in\omega) \delta_{nk} + 2g(\tilde{\phi})_l \delta_{n,k+l} \right]. \quad (46)$$

The gradient of the action takes the form

$$\begin{aligned} \frac{\partial S_{eff}}{\partial \tilde{\phi}_n} = & \beta \left[\{ (n\omega)^2 + 2g^2m^2 \} \tilde{\phi}_{-n} + g(\tilde{\phi})_{-n}^3 + 3g^2(\tilde{\phi}^5)_{-n} \right. \\ & \left. - Tr \left(\frac{\partial M}{\partial \tilde{\phi}_n} M^{-1} \right) \right]. \end{aligned} \quad (47)$$

3 Conclusions

We have looked at quantum harmonic oscillator in path integral formalism. We computed the partition function and correlation function for the harmonic oscillator.

We also looked at supersymmetric quantum mechanics in non-lattice formalism. We derived the bosonic and fermionic parts of the action for various superpotentials. We computed the fermionic operator in matrix form and gradients of the effective action for symmetric and non-symmetric superpotentials.

References

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